FINAL-YEAR PROJECT · SOFTWARE ENGINEERING · ARSD COLLEGE, UNIVERSITY OF DELHI

One tap, one incident.

A venue operations layer for crowd emergencies — because in a crowd you cannot hear, cannot be heard, and the network is jammed. *It escalates to 112; it does not replace it.*

SCOPE	STACK	STATUS
Single venue, multi-zone	FastAPI · PostGIS · React	Product name undecided

01

Problem statement

In a densely crowded venue, the established way to summon help — a voice call to an emergency number — fails at precisely the moment it is needed.

FAILURE ONE	FAILURE TWO	FAILURE THREE
Ambient noise makes speech unusable. You cannot hear the operator, and the operator cannot hear you.	Cell congestion drops calls when thousands of handsets contend for one tower — exactly when everybody needs it at once.	A call carries one person's account, once. Thirty callers describing one crush produce thirty conversations, not one incident.

India's Emergency Response Support System (112) handles the citizen-to-state path well, and this project does not attempt to replace it. But 112 routes a caller to a **state** control room. It does not give the operator of a particular venue a live picture of their own ground, it does not dispatch that venue's own marshals, medics or fire team, it does not merge thirty reports of one crush into a single prioritised incident, and it cannot push exit guidance to the people standing inside that venue.

The result is a gap. At a college fest, a stadium or a religious gathering, trained responders are frequently within a hundred metres of an incident, yet no system routes a report to them, resolves where that report came from, or tells the control room that thirty messages describe one event rather than thirty.

A second failure sits inside the same gap. Harassment at crowded events is common and heavily under-reported, and a visible or audible alert can increase the reporter's risk rather than reduce it. Existing tools alert either a state control room or a pool of unvetted volunteers — neither is the venue's own on-duty staff, and neither is discreet.

> *A crowded venue has responders, zones and an operations room, but no channel that turns what attendees can see into something that room can act on — quickly, without a voice call, without duplication, and without exposing a reporter who cannot afford to be seen.*

02

Description

The system is a web-based venue operations layer for crowd emergencies. Anyone inside a venue reports in one tap, optionally attaching a short description or a photograph. The system resolves the report to a venue zone, classifies it, merges it with other reports describing the same event, and presents the control room with a live, de-duplicated picture of its own ground. The operator dispatches the venue's own responders — the nearest available unit whose capability matches the incident — tracks acknowledgement and arrival, and broadcasts guidance to attendees where needed.

The central design premise is that **the network is unreliable exactly when the system matters**. A tap is a few hundred bytes and gets through when a voice call will not. Where location services fail, the reporter selects the nearest gate instead of being blocked. Where no trained classifier has scored an incident, the system reports the absence of a confidence value rather than inventing one.

Personal safety is handled as a first-class incident category rather than a separate application. A report of harassment can be submitted **discreetly** — no sound, no vibration, no visible change of screen state — and is matched to a responder by attribute as well as role, so a request needing a female responder is dispatched accordingly. Evidence is retained in an append-only record with a chain of custody, so a report can support a formal complaint later if the reporter chooses to make one.

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Objectives

01 Provide a single-tap, low-bandwidth reporting channel that functions without a voice call and degrades gracefully when GPS or network is unavailable.

02 Resolve every report to a venue zone, by device location where available and by reporter selection where it is not.

03 Merge independent reports describing the same event — matched on category, zone and time window — into one incident carrying a corroboration count.

04 Assign a severity tier to each incident, and expose the method that produced it, reporting no confidence value where no trained model has scored it.

05 Dispatch the nearest available responder whose role and attributes match the incident category, with a documented fallback when none matches.

06 Escalate unacknowledged high-severity incidents automatically on a timer, and provide a documented escalation path to 112.

07 Support discreet submission for personal-safety incidents, where a visible or audible alert would increase the reporter's risk.

08 Maintain an append-only audit trail per incident, recording every state change and every access to attached evidence.

09 Broadcast advisories and exit guidance from the control room to attendees.

04

Scope

IN SCOPE

- A single venue with multiple zones, configured by an administrator.
- Four actors: reporter (unauthenticated attendee), responder, control room operator, venue administrator.
- Categories: fire, medical, crowd, accident, violence, harassment, other.
- A mobile-first web application; no native app.
- Real-time updates to operator and responder screens without polling.

OUT OF SCOPE — STATED DELIBERATELY

- This is **not a replacement for 112**, and makes no claim to be.
- No API integration with ERSS — no public integration path is documented. Escalation is a specified handoff, not a backend connection.
- No hardware, wearables or panic buttons.
- No nationwide consumer deployment; scoped to a venue with its own responders on the system.
- No carrier-level handset location. A callback number lets an operator ring back, and is never used to locate.

05

Modules

Numbered to match the Level 1 data flow diagram, so the diagrams describe the real system rather than a parallel fiction.

#	MODULE	RESPONSIBILITY
1.0	**Report Intake**	Capture, zone resolution, discreet handling, photo attachment
2.0	**Triage**	Categorisation and severity assignment, with an honest confidence signal
3.0	**De-duplication**	Match and merge concurrent reports; maintain corroboration count
4.0	**Dispatch**	Match responder by role and attribute; nearest-available assignment
5.0	**Escalation**	Unacknowledged-incident timer, handoff to 112
6.0	**Broadcast**	Advisories and exit guidance to attendees
7.0	**Audit & Evidence**	Append-only incident record and chain of custody

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Tools and technologies

LAYER	CHOICE
Backend	Python 3.12, FastAPI
Database	PostgreSQL with PostGIS (geospatial) and pgvector (retrieval)
Migrations	Alembic
Cache & pub/sub	Redis
Real time	WebSockets
Frontend	React, mobile-first
Licence	MIT, self-hostable end to end

Every component is self-hosted. The system depends on no managed backend and no proprietary application builder — a deliberate change from the earlier prototype.

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Expected outcome

A working, self-hosted system demonstrating the full path from report to resolution: an attendee submits a report, it is zone-resolved and triaged, duplicate reports collapse into one incident with a visible corroboration count, the control room dispatches a matched responder, the responder acknowledges and resolves, and the entire sequence is recoverable from the audit trail.

Evaluated by a **structured drill** — scripted scenarios run with volunteers, with response times and de-duplication accuracy measured and reported as numbers rather than claims. Accompanied by the required documentation: data flow diagrams at Levels 0, 1 and 2, and a use case diagram with written case descriptions.

08

Prior work by the team

An earlier prototype was built for the Cognitive Chaos 2026 hackathon and reached the finale. It established the reporting flow and the incident category set, both of which carry forward. It ran on a managed backend and a proprietary app builder required by the competition, used hardcoded data for a small number of zones, and had no trained model behind its triage.

The present work is an independent rebuild that removes those dependencies and implements the logic the prototype only demonstrated. Citing this honestly is a strength: the repositioning away from "a 112 replacement" came directly out of judge feedback, and the current framing is the result.

> **OPEN DECISION** The product name is not settled. Everywhere a name is needed, the placeholder <NAME> is used. The submission title should read "<NAME>: A Venue Operations Layer for Crowd Emergency Response and Responder Dispatch." Renaming is cheapest now — there are no users, no release and no migrations depending on it.